Product, Quotient, and Trigonometric Derivatives

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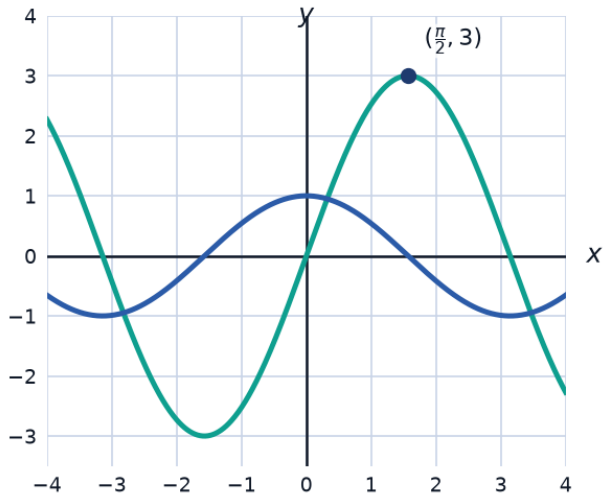
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The Product Rule**

A differentiable function $f(t)$ is represented by the graph below. What is the total change in $f(t)$ from $t = 2$ to $t = 5$?

2 **[Foundational]** • **Target: The Quotient Rule**

Given $\int_0^1 1x^2\,dx = \frac{1}{3}$ and $\int_1^3 1x^2\,dx = \frac{26}{3}$, find $\int_0^3 1x^2\,dx$.



3 **[Foundational]** • **Target: Derivatives of Basic Trigonometric Functions**

Consider the function $f(x) = 3x^2 - 6x + 5$. What is the instantaneous rate of change of $f(x)$ at $x = 2$?

4 **[Foundational]** • **Target: Higher-Order Derivatives & Acceleration**

A conical tank has height 15 ft and radius 7 ft at the top (radius:height ratio constant as it drains). Water drains at 6 ft^3/min. Find $\frac{dh}{dt}$ (in ft/min) when the water depth is 13 ft.

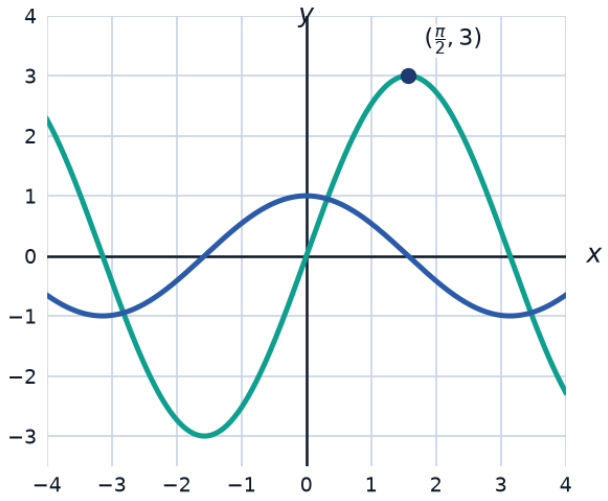
5 **[Foundational]** • **Target: The Product Rule**

Product rule for $\frac{d}{dx}[f(x)g(x)]$:.

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6 [Foundational] • Target: The Quotient Rule

Use the quotient rule to find $f'(x)$ if $f(x) = \frac{x}{4x+2}$.



7 [Foundational] • Target: Derivatives of Basic Trigonometric Functions

Use the quotient rule to find $f'(x)$ if $f(x) = \frac{x}{5x+3}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Higher-Order Derivatives & Acceleration

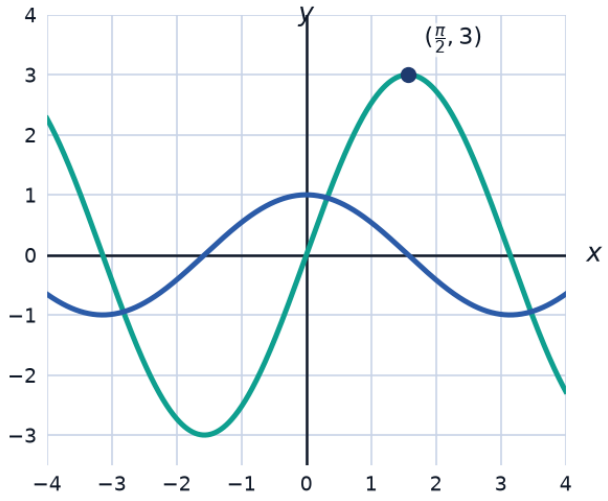
Use the quotient rule to find $f'(x)$ if $f(x) = \frac{x}{2x+2}$.

9 [Intermediate] • Target: The Product Rule

Use the product rule to find $f'(x)$ if $f(x) = x^4(3x - 4)$.

10 [Intermediate] • Target: The Quotient Rule

Use the product rule to find $f'(x)$ if $f(x) = x^4(2x + 3)$.



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11 [Intermediate] • Target: Derivatives of Basic Trigonometric Functions

Use the product rule to find $f'(x)$ if $f(x) = x^4(4x - 4)$.

12 [Intermediate] • Target: Higher-Order Derivatives & Acceleration

Use the product rule to find $f'(x)$ if $f(x) = x^3(5x - 1)$.

13 [Intermediate] • Target: The Product Rule

Use the product rule to find $f'(x)$ if $f(x) = x^3(5x + 3)$.

14 [Intermediate] • Target: The Quotient Rule

Use the product rule to find $f'(x)$ if $f(x) = x^3(4x - 1)$.

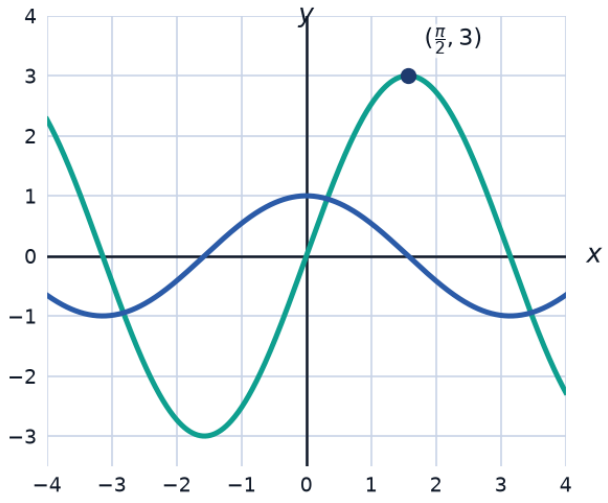
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Derivatives of Basic Trigonometric Functions • Cross-Lesson: 3.01 The Chain Rule

Use the product rule to find $f'(x)$ if $f(x) = x^4(4x + 1)$.

16 [Challenging] • Target: Higher-Order Derivatives & Acceleration • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Use the product rule to find $f'(x)$ if $f(x) = x^4(5x - 1)$.



17 [Challenging] • Target: The Product Rule • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Use the product rule to find $f'(x)$ if $f(x) = x^3(2x + 1)$.

18 [Challenging] • Target: The Quotient Rule • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Use the product rule to find $f'(x)$ if $f(x) = x^3(1x + 2)$.

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19 [Challenging] • Target: Derivatives of Basic Trigonometric Functions • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Use the product rule to find $f'(x)$ if $f(x) = x^3(2x + 3)$.

20 [Challenging] • Target: Higher-Order Derivatives & Acceleration • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Use the product rule to find $f'(x)$ if $f(x) = x^2(4x + 2)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Product Rule

Find the derivative of $f(x) = x^3 \sin x$.

a. $3x^2 \sin x + x^3 \cos x$

b. $3x^2 \cos x$

c. $3x^2 \sin x - x^3 \cos x$

d. $x^2(3 \sin x + \cos x)$

22 [Foundational] • Target: The Product Rule

If $h(x) = f(x)g(x)$ with $f(2) = 3, f'(2) = -1, g(2) = 4,$ and $g'(2) = 5$, what is $h'(2)$?

a. 11

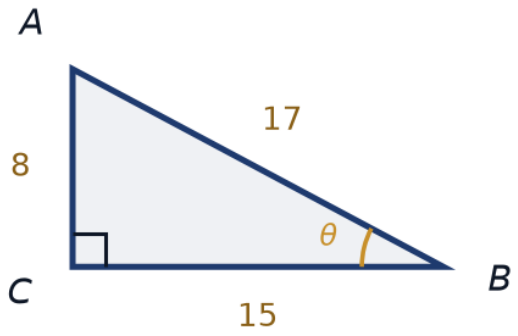
b. 19

c. -14

d. 15

23 [Foundational] • Target: The Product Rule

Find $\frac{d}{dx}[e^x \cos x]$.



a. $e^x(\cos x - \sin x)$

b. $e^x(\cos x + \sin x)$

c. $-e^x \sin x$

d. $e^x \cos x$

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24 [Foundational] • Target: The Quotient Rule

Differentiate $y = \frac{2x-3}{x+4}$.

a. $\frac{11}{(x+4)^2}$

b. $\frac{5}{(x+4)^2}$

c. $\frac{2x+5}{(x+4)^2}$

d. $\frac{2}{(x+4)^2}$

25 [Foundational] • Target: The Quotient Rule

Let $k(x) = \frac{f(x)}{g(x)}$. If $f(1) = 6, f'(1) = 2, g(1) = 3,$ and $g'(1) = -1$, find $k'(1)$.

a. $\frac{4}{3}$

b. 0

c. 2

d. $-\frac{4}{3}$

26 [Foundational] • Target: The Quotient Rule

Find the derivative of $f(x) = \frac{e^x}{x^2}$.

a. $\frac{e^x(x-2)}{x^3}$

b. $\frac{e^x(x+2)}{x^3}$

c. $\frac{e^x}{2x}$

d. $\frac{e^x(x-1)}{x^3}$

27 [Foundational] • Target: Derivatives of Basic Trigonometric Functions

What is $\frac{d}{dx}[\tan x]$?

a. $\sec^2 x$

b. $\sec x \tan x$

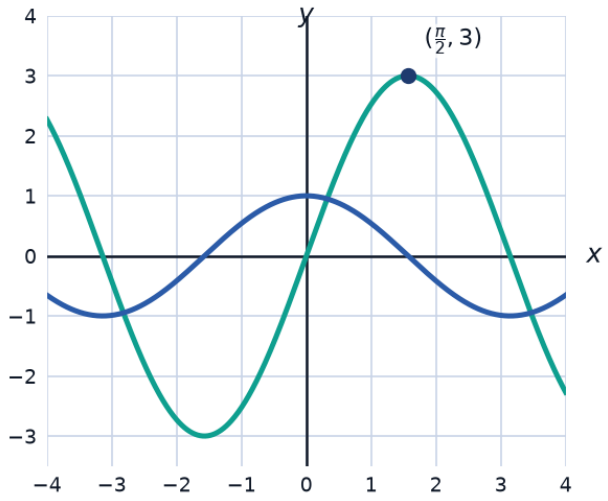
c. $-\csc^2 x$

d. $\cot x$

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28 [Foundational] • Target: Derivatives of Basic Trigonometric Functions

What is $\frac{d}{dx}[\sec x]$?



a. $\sec x \tan x$

b. $\sec^2 x$

c. $-\csc x \cot x$

d. $\tan^2 x$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Derivatives of Basic Trigonometric Functions

What is $\frac{d}{dx}[\cot x]$?

a. $-\csc^2 x$

b. $\csc^2 x$

c. $-\sec^2 x$

d. $-\csc x \cot x$

30 [Intermediate] • Target: Higher-Order Derivatives & Acceleration

Find the second derivative $f''(x)$ of $f(x) = \sin x$.

a. $-\sin x$

b. $\sin x$

c. $-\cos x$

d. $\cos x$

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31 [Intermediate] • Target: Higher-Order Derivatives & Acceleration

If position is $s(t) = t^4 - 2t^3 + 5$, what is the acceleration $a(t)$ at $t = 2$?

a. 24

b. 36

c. 12

d. 48

32 [Intermediate] • Target: Higher-Order Derivatives & Acceleration

What is the 50th derivative of $f(x) = \cos x$?

a. $-\cos x$

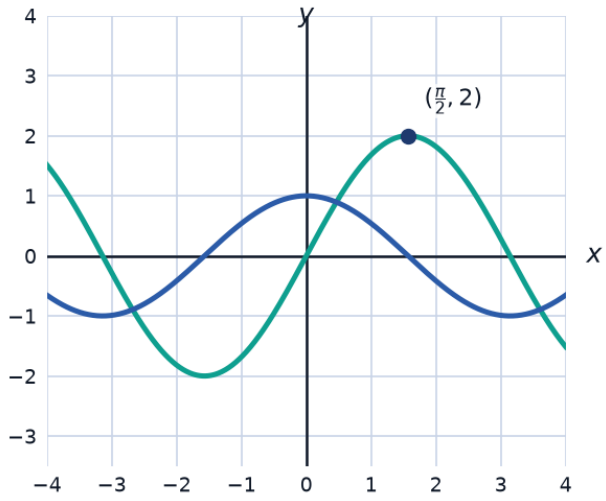
b. $\cos x$

c. $\sin x$

d. $-\sin x$

33 [Intermediate] • Target: The Product Rule

Use the product rule to find $f'(x)$ if $f(x) = x^2(2x - 1)$.



a. $2x(3x - 1)$

b. $4x$

c. $6x^2 - 2x - 2$

d. $x(6x - 1)$

34 [Intermediate] • Target: The Quotient Rule

Use the product rule to find $f'(x)$ if $f(x) = x^2(1x + 3)$.

a. $2x$

b. $x(3x + 7)$

c. $3x(x + 2)$

d. $3x^2 + 6x - 1$

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35 [Intermediate] • Target: Derivatives of Basic Trigonometric Functions

Use the quotient rule to find $f'(x)$ if $f(x) = \frac{x}{2x+0}$.

a. 2

b. $\frac{1}{2x}$

c. 0

d. 1

36 [Intermediate] • Target: Higher-Order Derivatives & Acceleration

Use the chain rule to find $f'(x)$ if $f(x) = (-3x-3)^4$.

a. $4(-3x-3)^3$

b. $-12(-3x-3)^4$

c. $-12(-3x-3)^3$

d. $-3(-3x-3)^3$

37 [Intermediate] • Target: The Product Rule

Use the chain rule to find $f'(x)$ if $f(x) = (-2x-5)^2$.

a. $2(-2x-5)^1$

b. $-4(-2x-5)^2$

c. $-2(-2x-5)^1$

d. $-4(-2x-5)^1$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Quotient Rule • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Use the chain rule to find $f'(x)$ if $f(x) = (-2x-5)^3$.

a. $3(-2x-5)^2$

b. $-2(-2x-5)^2$

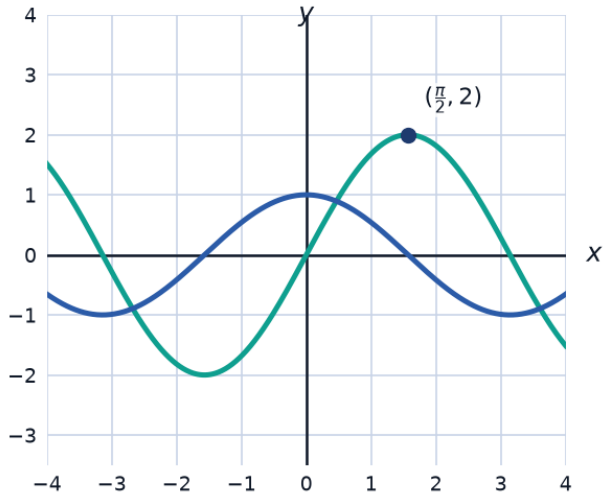
c. $-6(-2x-5)^3$

d. $-6(-2x-5)^2$

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39 [Challenging] • Target: Derivatives of Basic Trigonometric Functions • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Use the chain rule to find $f'(x)$ if $f(x) = (-2x - 6)^4$.



a. $-8(-2x - 6)^4$

b. $4(-2x - 6)^3$

c. $-2(-2x - 6)^3$

d. $-8(-2x - 6)^3$

40 [Challenging] • Target: Higher-Order Derivatives & Acceleration • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Use the chain rule to find $f'(x)$ if $f(x) = (-3x + 2)^3$.

a. $-9(-3x + 2)^3$

b. $-9(-3x + 2)^2$

c. $3(-3x + 2)^2$

d. $-3(-3x + 2)^2$

41 [Challenging] • Target: The Product Rule • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Use the chain rule to find $f'(x)$ if $f(x) = (-3x + 3)^3$.

a. $3(-3x + 3)^2$

b. $-9(-3x + 3)^2$

c. $-9(-3x + 3)^3$

d. $-3(-3x + 3)^2$

42 [Challenging] • Target: The Quotient Rule • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Use the chain rule to find $f'(x)$ if $f(x) = (-2x - 1)^3$.

a. $3(-2x - 1)^2$

b. $-2(-2x - 1)^2$

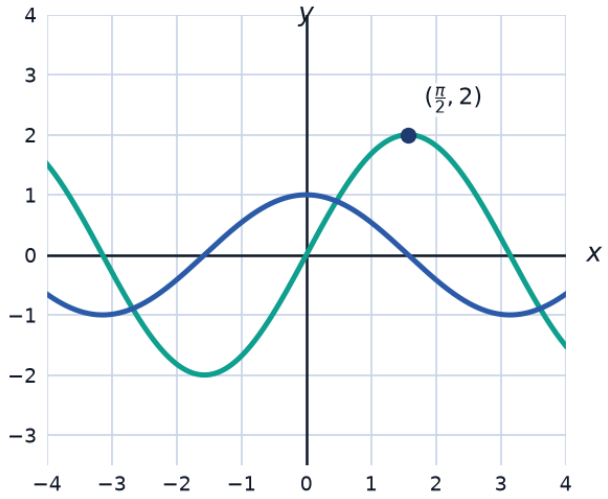
c. $-6(-2x - 1)^2$

d. $-6(-2x - 1)^3$

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43 [Challenging] • Target: Derivatives of Basic Trigonometric Functions • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Use the chain rule to find $f'(x)$ if $f(x) = (3x - 1)^4$.



a. $4(3x - 1)^3$

b. $12(3x - 1)^3$

c. $12(3x - 1)^4$

d. $3(3x - 1)^3$

44 [Challenging] • Target: Higher-Order Derivatives & Acceleration • Cross-Lesson: 8.01 Area Between Curves

Use the chain rule to find $f'(x)$ if $f(x) = (3x - 2)^4$.

a. $12(3x - 2)^4$

b. $4(3x - 2)^3$

c. $3(3x - 2)^3$

d. $12(3x - 2)^3$

45 [Challenging] • Target: The Product Rule • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Use the chain rule to find $f'(x)$ if $f(x) = (3x + 4)^3$.

a. $9(3x + 4)^3$

b. $9(3x + 4)^4$

c. $9(3x + 4)^2$

d. $3(3x + 4)^2$

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